

Year 8 Science Investigating a Pendulum

INVESTIGATION 1.2

The period of a pendulum

AIM To investigate the effects of mass and length on the period of a pendulum

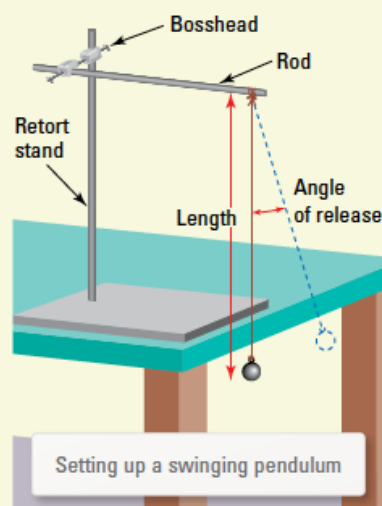
Materials:

length of string (at least 80 cm long)
set of slotted masses
retort stand with bosshead
pair of scissors and a one-metre ruler
stopwatch or clock with a second hand

METHOD AND RESULTS

Part 1: The effect of mass

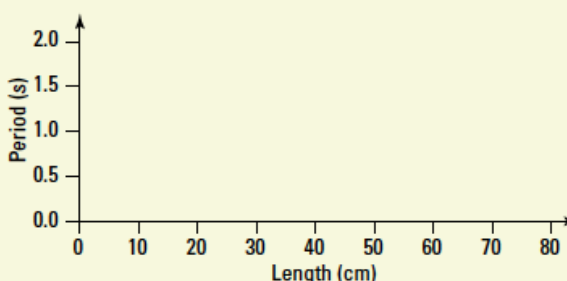
- ▶ Set up your pendulum so it can swing freely. Start with the largest possible length and the smallest weight.
- 1 Copy the table below right into your logbook, and record the mass and the length of the pendulum in it. The length should be measured from the top of the pendulum to the bottom of the swinging mass, as shown in the diagram.
- ▶ Pull the mass aside so that the angle of release is about 20° . Take note of the height from which the mass is released so that this angle of release is used throughout the experiment.
- ▶ Release the pendulum. Measure the time taken for 10 complete swings of the pendulum. Repeat your measurement at least twice to find the average time for 10 swings. Use this average to calculate the time taken for one complete swing (the period).



- 2 Record all the measurements in your table.
- ▶ Repeat this procedure for three larger masses, completing the table as you go.

Part 2: The effect of length

- 3 Construct a table like the one on page 13 to identify all of the variables that need to be considered for an investigation of the effect of length on the period of a pendulum.
- 4 Construct a second table in which to record your measurements. Remember that this time you'll be testing four different lengths without changing the mass. Use the same procedure as you did in part 1 for measuring the period.
- 5 Draw a line graph to show how the period of the pendulum is affected by its length.



DISCUSS AND EXPLAIN

- 6 How does the mass of the pendulum affect its period?
- 7 How does the length of the pendulum affect its period?
- 8 The period of most standard clock pendulums is one second. Use your graph to predict the length of a standard clock pendulum.
- 9 Explain why it is a good idea to measure the time for 10 swings rather than just one.

The effect of mass on the period of a pendulum

Length of pendulum = _____ cm

Angle of release = 20°

Mass (grams)	Time taken for 10 complete swings (seconds)				Period (seconds)
	Trial 1	Trial 2	Trial 3	Average	